De novo transcriptome assembly and annotation for gene discovery in *Salamandra salamandra* at the larval stage

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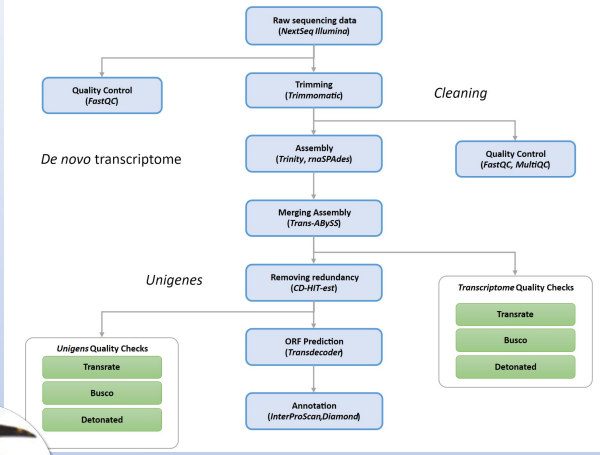
Background

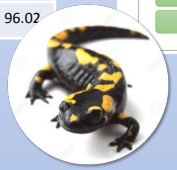
Ten larvae of the *Salamandra salamandra* species were selected representing two distinct behavioural profiles: slow, less active and less exploratory behaviour, versus fast, more active and more exploratory behaviour. The brains were dissected and immediately preserved in RNAprotect Tissue Reagent until RNA extraction.

Sample code	Phenotype	Raw sequences	% Trimmed reads
R530	Slow	148383674	96.57
R531	fast	113170212	96.18
R538	slow	131207312	96.48
R541	fast	101470648	96.27
R547	fast	116726758	96.64
R560	slow	123732768	96.51
R564	slow	99188392	96.53
R565	fast	98856326	96.68
R572	fast	117839158	96.43
R573	slow	102857670	96.02



Methods

De novo transcriptome assembly involves: first a trimming phase, followed by a quality control step. Use of two different assemblers and subsequent transcriptome validation. A merge phase to get a unique assembly consensus file and a step of removing transcript redundancies. Finally ORF prediction and annotation (both by homology and functional) of the transcriptome is applied.



	Trinity	rnaSPAdes	CD-HIT-est (Unigens)
TransRate v.1.0.3			
Total transcripts	1207872	1094271	1146571
N50	1742	1979	1529
GC content (%)	45.00	45.00	44.73
BUSCO v.3.0.2			
Complete BUSCOs (C)	254 (99.2%)	252 (98.8 %)	249 (97.6%)
Fragmented BUSCOs (F)	0 (0.0 %)	1 (0.4 %)	5 (2.0 %)
Missing BUSCOs (M)	1 (0.8%)	2 (0.8 %)	1 (0.4 %)

Assembly results:

From the data of the larval brain samples, we obtained 1,153,432,918 pairs of trimmed reads. Running rnaSPAdes generated a total of 1,094,271 transcripts. running Trinity a total of 1,207,872 transcripts.
The final assembled transcriptome comprised a total of 1146571 transcripts with an N50 of 1529 bp. The completeness for BUSCO evaluation was above 97.6%.

Annotation results:

The contigs were aligned with DIAMOND on NCBI nr, Swiss-Prot and TrEMBL to retrieve the best matching annotations. As a result of blastx analysis we obtain: 220041 sequences of 49.86%, 154324 (34.97%), 220521 (49.97%), respectively.

Annotation statistics	
Number of blastx results	
NCBI nr	220041 (49.86 %)
Swiss-Prot	154324 (34.97 %)
TrEMBL	220521 (49.97 %)